Samantha Burd

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EDUCATION

University of California, Berkeley Berkeley, CA
B.A Astrophysics and Economics Expected Graduation: May 2028
Relevant Coursework: Introduction to Computer Programming and Numerical Methods, Mechanics and Wave Motion, Heat, Electricity, and Magnetism, Three-Dimensional Modeling for Design, NewSpace@Berkeley DeCal, Python for Astronomers

PROFESSIONAL EXPERIENCE

Glenair Glendale, CA
Materials Engineering Intern May 2026 - August 2026

- Diagnosed a defect as human error after systematically reducing the potential process failure points from nine to one, compiling my findings in a comprehensive deck-style report presented to 100+ senior leadership staff and interns
- Identified human error as the root cause of electroless-nickel plating defects by narrowing 9 potential process failure points to 1 through design-of-experiments and coupon analysis and presented findings to 100+ leaders and interns
- Identified surface contaminants on EN plated shells utilizing materials characterization techniques such as Scanning Electron Microscopy, Electron Dispersive Spectroscopy, and Keyence Microscopy up to 3000x magnification.

Astroscale U.S. Berkeley, CA
Policy Strategy Consultant (Contract) January 2026 - May 2026

- Projected a 33% reduction in Active Debris Removal (ADR) licensing timelines for commercial operators by developing a three-phase policy strategy around consolidated authorization, legislative reform, and standardized regulatory language
- Identified a principal-agent dilemma restricting commercial ADR growth, driven by fragmented oversight across four federal agencies (NASA, FCC, FAA, NOAA) and a resulting free-rider problem resulting in disincentivized investment
- Designed a tiered enforcement structure to deter free-riding and protect commercial ADR investment, scaling fines from $150K to $10M by violation severity and benchmarking against FCC non-compliance precedent and industry standards

Space Enterprise at Berkeley Berkeley, CA
Structures Member January 2026 - June 2026

- Fabricated a composite rocket nose cone through hands-on manufacturing, performing the cutting, orientation, and layup of fiberglass layers while maintaining precise resin-to-fiber ratios to satisfy structural and aerodynamic requirements
- Applied hand-finishing techniques including sanding, machining, and surface preparation to achieve precise tolerances on composite and metal components, ensuring structural integrity and fit for flight-ready hardware

RESEARCH EXPERIENCE

VORTEX – Visualization of Relativistic Tidal EXtremes Santa Clarita, CA
Mechanical Team Member June 2026 - Present

- Identified a hollow-stem plunger as the optimal mechanism for dynamically controlling drain radius in an analog black-hole apparatus governed by the Vaidya metric, after evaluating iris apertures, butterfly valves, and other designs
- Produced three iterations of a hollow-stem plunger design, utilizing SolidWorks to model and evaluate the feasibility of various cone-angle designs and radius configurations, coordinating with the electrical team to ensure design compatibility

SKILLS AND INTERESTS

Technical Skills: Python, NumPy, Matplotlib, SolidWorks, OnShape
Materials Characterization: Scanning Electron Microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDS), Keyence Microscopy
Interests: Cosmology, Dark Matter, Research, Entrepreneurship, Commercial Space, Photography, Video Editing